HOLLYWOOD DATA ANALYSIS REPORT

Power BI | Power Query | Hollywood Grossing Movies Dataset

Analysis of top-grossing films across multiple decades, covering worldwide sales, budgets, distributors, genres, and runtime trends.
Data Source: https://www.kaggle.com/datasets/sanjeetsinghnaik/top-1000-highest-grossing-movies

1. Objective

The objective of this report is to analyze the highest-grossing Hollywood movies using Power BI and Power Query, and to present a high-level KPI dashboard for studio executives, entertainment analysts, and financial stakeholders. The report provides visibility into global box-office performance, distributor dominance, genre trends, budget patterns, runtime distribution, and billion-dollar movie behavior. The analysis leverages a curated dataset of top-performing films across multiple years to surface actionable insights and support strategic decision-making in production planning, marketing, and distribution.

2. Data Dictionary

The dataset contains detailed movie-level information including financial performance, production details, runtime, genre, and distributor metadata.

2.1 Core Fields

Field Name	Description
Title	Movie title
Year	Release year of the movie
Distributor	Studio or company responsible for distribution
Budget (in $)	Total production budget
Domestic Opening (in $)	Revenue generated during opening weekend in the U.S.
Domestic Sales (in $)	Total U.S. box-office revenue
International Sales (in $)	Total non-U.S. box-office revenue
Worldwide Sales (in $)	Total global box-office revenue
Genre	Primary and secondary genres
Running Time	Total runtime of the movie
License	Rating (G, PG, PG-13, R)
Movie Info	Short plot description

3. Dataset Overview

The dataset includes top-grossing Hollywood films spanning multiple decades. It captures:

- Financial performance (domestic, international, worldwide)
- Production budgets
- Distributor influence
- Genre distribution
- Runtime categories
- Billion-dollar movie trends
- Year-wise movie counts

This enables comprehensive analysis of box-office behavior and long-term industry patterns.

4. Data Preparation (Power Query)

Power Query was used to clean and transform the dataset:

- Promoted headers
- Changed data types
- Replaced errors
- Renamed columns
- Removed unnecessary fields
- Created runtime groups (e.g., `<2 hr`, `2–2.5 hr`, `2.5–3 hr`, `>3 hr`)
- Standardized distributor names
- Ensured numeric fields were properly typed for aggregation

These steps produced an analysis-ready model for the Hollywood Dashboard.

5. Dashboard Insights

5.1 Highest Grossing Movies (Worldwide)

Top global performers include:

- Avatar – $2.928B
- Avengers: Endgame – $2.797B
- Avatar: The Way of Water – $2.310B
- Star Wars: The Force Awakens – $2.072B
- Avengers: Infinity War – $2.049B

These films dominate due to strong international markets, franchise power, and high production value.

5.2 Domestic vs International Sales

Movie	Domestic ($B)	International ($B)	Worldwide ($B)
Avatar	0.76	2.16	2.92
Avengers: Endgame	0.85	1.93	2.79
Avatar: The Way of Water	0.68	1.62	2.31

Insight: International markets often contribute 60–75% of total revenue, emphasizing the importance of global distribution strategies.

5.3 Budget Trends

Average budget across all films is approximately **$97M**. High-budget films (>$200M) tend to dominate worldwide sales due to:

- Large-scale production
- Heavy VFX usage
- Global marketing campaigns
- Franchise-based storytelling

5.4 Runtime Analysis

Runtime Group	Count
< 2 hr	41
2–2.5 hr	29
2.5–3 hr	21
> 3 hr	7

Insight: Most billion-dollar films fall between **2–3 hours**, aligning with modern blockbuster pacing.

5.5 Genre & Distributor Performance

Top Genres:

- Adventure
- Action
- Sci-Fi
- Fantasy

Top Distributors:

- Walt Disney Studios Motion Pictures
- Universal Pictures
- Warner Bros
- 20th Century Fox

Disney leads worldwide due to Marvel, Star Wars, and animated franchises.

5.6 BillionDollar Movies

Recent billion-dollar films include:

- The Super Mario Bros. Movie (2023)
- Jurassic World Dominion (2022)
- Avatar: The Way of Water (2022)
- Spider-Man: No Way Home (2021)
- Frozen II (2019)
- Aquaman (2018)

Franchise films and sequels dominate the billion-dollar list.

5.7 Year-Wise Trends

Year-wise analysis shows:

- Peaks around 2015 and 2019
- Declines during pandemic years
- Recovery in 2022–2023

This aligns with industry-wide production cycles and global events.

6. Future Scope

Initiative	Description
Predictive Box-Office Modeling	Forecast worldwide sales using budget, genre, distributor, and franchise data.
Genre Profitability Analysis	Identify which genres deliver the highest ROI relative to budget.
Distributor Market Share Tracking	Analyze long-term shifts in distributor dominance across decades.
Franchise Performance Insights	Evaluate multifilm franchises to optimize sequel timing and investment.

7. Conclusion

This Hollywood Analytics Report provides a comprehensive view of global movie performance, highlighting the financial, structural, and creative factors that drive blockbuster success. The insights support strategic planning for studios, distributors, and investors, enabling data-driven decisions in production, marketing, and franchise development.